

width in comparison to human-to-human communication (which needs high-quality source and higher bandwidth).

Traditionally, existing technologies like Wi-Fi and GSM are used to enable M2M communication. These technologies, centered around high bandwidth, are overkill for M2M. They also consume excessive power and are therefore energy-inefficient in battery-powered applications.

Therefore, in recent years, new standards were developed for wide-area, radio-based and low-power technologies that are optimised for low data rates. Usually known as low-power WAN (LPWAN), these are highly power efficient. LoRaWAN and NB-IoT are two such competing standards.

LoRaWAN is already being used in some parts of the world. The general use cases of LoRaWAN are in agriculture, where it is used to establish communication links with sensors installed on the field. In a similar way, LoRaWAN can be deployed for public utilities like parking systems in urban areas. Telcos can tap into LoRaWAN by providing LoRaWAN services to businesses.

Offering value-added and intelligent services

Merging of virtual and physical worlds. Finding our way from physical to virtual world and vice-versa is a concept similar to physical web proposed by Google. But, this approach of Google works in a personal-area network (PAN) space.

Proximity sensing (ProSe) is based on standards of a service on top of 4G LTE cellular service that enables two devices to detect their proximity and exchange data. Some of the practical use cases are mentioned below.

Finding people. Suppose you are attending a conference and want to know whether anyone from your LinkedIn network is present there. If you and your links have ProSe installed on your smartphones, LinkedIn app will scan the nearby people quickly and pop up on your phone's screen the list of people from your LinkedIn network.

Finding places. Suppose you are in a new city and you must locate the nearest train station or bus stand to travel to another part of the city. If every train, bus and public transport aggregation point in the city consists of a ProSe enabled device, these can sync up with the ProSe service of your smartphone to present you with an interactive application that helps you find your way to the closest point. Additionally, if you are able to communicate with those devices, they can present you with information like train/bus timings.

To sum up

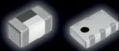
Analytics, control and monitoring make up the highly obvious IoT use cases. Many jobs are also being generated due to innovative applications of the IoT. Many beginners and experienced professionals are taking IoT training and certification as it promises a lucrative career. A new wave of collaborative applications is merging the physical world into our virtual lives, allowing us to make a transition from physical to virtual, or virtual to physical world, seamlessly. **EFY**

Attracting Tomorrow



TDK Technology Linking the world.

Thin-film and multilayer RF chip components



Miniaturized pressure sensors for highest precision

Power management modules with high efficiency based on SESUB embedding



Advanced lens actuators contribute to higher image quality

<https://product.tdk.com>